

Securing a Wireless Network

Cybersecurity Portfolio Project

1. Project Overview

This project focused on improving wireless network security using router configuration settings. The lab demonstrated how wireless encryption, authentication methods, and guest network controls help protect Wi-Fi networks from unauthorized access.

2. Objective

The goal of this project was to configure secure wireless network settings by reviewing Wi-Fi encryption options, strengthening authentication methods, and managing guest network access.

3. Tools Used

- TP-Link Router Interface
- Wireless Router Configuration Panel
- Wi-Fi Security Settings
- Virtual Lab Environment

4. Skills Demonstrated

- Wireless Network Security
- Router Configuration
- Wi-Fi Encryption
- Access Control
- Guest Network Management
- Security Hardening
- Network Administration

5. Lab Environment

The lab environment used a TP-Link wireless router management interface to configure and review wireless security settings for both standard and guest wireless networks.

6. Steps Performed

1. Logged into the wireless router administration interface.
2. Navigated to Wireless Settings.
3. Reviewed available wireless security options.
4. Configured WPA/WPA2 wireless encryption settings.
5. Examined WPA2/WPA3 security options.
6. Reviewed guest network configuration settings.
7. Configured guest wireless access options.
8. Documented wireless security configurations.

7. Screenshots

Figure 1 — Configuring Wireless Security Settings

- Reviewed available wireless security protocols:
 - WPA/WPA2-Personal
 - WPA2/WPA3-Personal
 - WPA/WPA2-Enterprise
- Examined wireless encryption configuration settings.

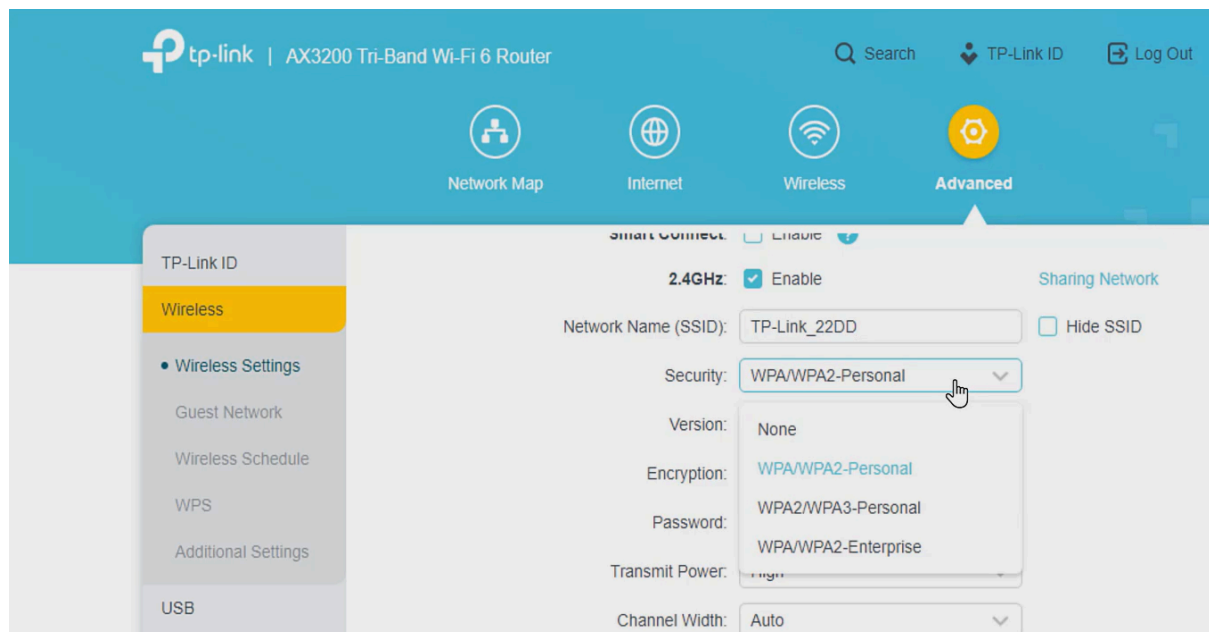
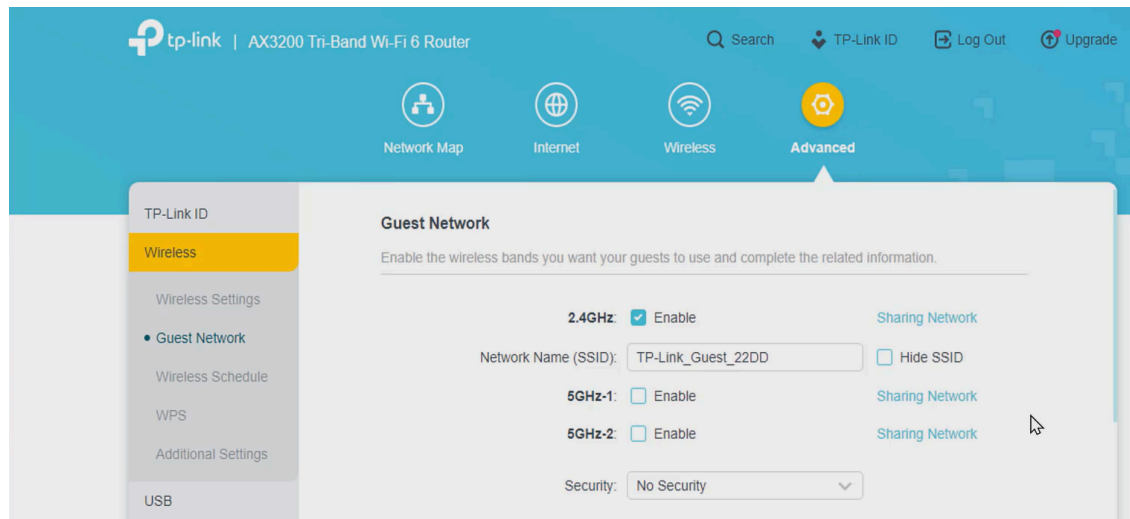


Figure 2 — Guest Network Configuration

- Reviewed guest network settings.
- Configured guest wireless access controls.
- Examined guest SSID and security settings.



8. Security Benefits

Wireless security configuration helps:

- Prevent unauthorized Wi-Fi access
- Protect network traffic
- Secure wireless communications
- Isolate guest users
- Reduce wireless attack risks

Using stronger encryption methods such as WPA2 and WPA3 improves protection against wireless attacks and unauthorized access attempts.

9. What I Learned

I learned how wireless routers use encryption and authentication settings to secure Wi-Fi networks and control access for connected devices.

I also gained experience configuring guest networks and understanding how wireless security settings help protect home and enterprise environments.

10. Conclusion

This project demonstrated practical cybersecurity skills related to wireless network security, router configuration, and Wi-Fi protection using modern wireless security settings and guest network controls.